

SHREY BANSAL

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Junior Undergraduate, Computer Science and Engineering

EDUCATION

Indian Institute of Technology, Kanpur	📅 2021 - 2025
Computer Science and Engineering	CPI: 8.92 / 10
Green Valley High School	📅 2019 - 2021
CBSE Class XII	Grade: 96.6%
Navrachana School Sama	📅 2018 - 2019
CBSE Class X	Grade: 98.2%

PROJECTS

E-Vaidya : HC Automation 📅 Mar'23 - Apr'23
Course Project, Prof. Indranil Saha 🌐 Project

- Automated appointment procedures at IITK Health Center using a **web-based app** with separate interfaces for different user roles
- Implemented features like appointment scheduling, medicine prescription, doctor's list scheduling, **pipeline to conduct appointments**
- Utilized technologies like **ReactJS, ExpressJS, NodeJS, MongoDB, JWT Tokens, and bcrypt** for efficient and secure development

CSE - BUBBLE 📅 Mar'23 - Apr'23
Course Project, Prof. Urbi Chatterjee 🌐 Project

- Used **Verilog HDL** to design a **processor** with a **custom Instruction Set Architecture**, taking reference from **MIPS-32 ISA**
- Designed op-code formats and modules for efficient **instruction handling**, encompassing **ALU operations and branching mechanisms**
- Converted the **Assembly code to Machine Code** to perform **Bubble Sort Algorithm** using the processor

Space Travel System 📅 Mar'23 - Mar'23
Course Project, Prof. Indranil Saha 🌐 Project

- Developed an **interactive command-line program** in C++ using **OOPS concepts** and role-based system offering 3 roles to the users
- Designed and made a **comprehensive ticketing system** with features for booking, viewing, and deleting tickets
- Ensured code integrity and maintainability by employing **data encapsulation and modular code organization**

Neural Machine Translation 📅 Dec'22 - Feb'23
Brain and Cognitive Society, IIT Kanpur 🌐 Project

- Developed **Neural Machine Translation** model from scratch, for **English-Hindi conversion** achieving **92% accuracy** on Kaggle dataset
- Made a **LSTM Seq2Seq** model with **encoder-decoder architecture**
- Employed **tokenization techniques** to preprocess input sentences and **generated embeddings** using TensorFlow embeddings, enhancing the model's ability to handle **complex linguistic structures**

Tripsay 📅 May'23 - May'23
Self Project 🌐 Project

- Developed **travel expert AI** using **OpenAI API** endpoints in **NextJS**
- Enforced a user-friendly interface supporting **text and audio input**
- Created chatbot capable of **natural language conversations**, allowing users to query itineraries, generate personalized itineraries, etc.
- Utilized **Typescript** for enhanced reliability, and type safety

RESPONSIBILITIES

Technical Secretary 📅 Feb'23 - Present
ACA, IIT Kanpur

- Organized and planned annual activities (Happy Hour and Farewell) conducted by ACA

Events Secretary 📅 Jun'22 - Apr'23
Entrepreneurship Cell, IIT Kanpur

- Organized the Annual Flagship event E-Summit'22 with a team of 20 secretaries
- Hosted and managed the different competitions held by E-Cell
- Lead a team of 20 volunteers to structure, publicize, and organize entrepreneurial events on campus

WORK EXPERIENCE

Vibinex 📅 Jun'23 - Present
Software Development Internship

- Executed a **CI/CD pipeline** to automate **deployment process** of the chrome extension, utilizing **GitHub Workflows** to trigger pipeline on **pull request merge**
- Developing feature in **rust** to assess **impact of pull request changes** on a codebase's **dependency graph**, enabling precise **evaluation of relevance** for a specific user

NOCIW 📅 May'23 - Present
Full-Stack Android Development Internship

- Implemented feature to get **location** of nearby people with same interests on **React Native App** and display results, categorized based on interest
- Designed and upgraded various features to display local business' information on the **community platform**
- Made the feature for users to **claim rewards** based on their collected points, as well as view their upcoming rewards
- Building a feature to record voice messages in chat and store the **base64 string** on **AWS S3 Bucket** and display the waveform

ACHIEVEMENTS

- Received the **Academic Excellence Award**, IIT Kanpur for two consecutive times for outstanding academic performance
- Secured an **All India Rank 174** in Joint Entrance Examination (**JEE Advanced 2021**) among 142k students
- Secured an **All India Rank 238** in Joint Entrance Examination (**JEE Main 2021**) among 939k students
- KVPY Fellowship**, securing **All India Rank 342 (SX)** and **511 (SA)** conducted by **IISc Bangalore**
- Among **Top 1%** in State in **IOQP** - conducted by **IAPT** in 2021.

SKILLS

- Languages** - C, C++, Rust, Python, JavaScript, Bash, Verilog HDL, Assembly Language, MIPS
- Development** - Typescript, React.js, Next.js, Node.js, Express.js, MongoDB, React Native, HTML
- Libraries** - NumPy, Pandas, Matplotlib, TensorFlow, Keras, Seaborn, OpenAI
- Tools** - $\LaTeX$, PostgreSQL, Tailwind CSS, Autodesk Inventor, Bootstrap, Git, Ghidra

RELEVANT COURSES

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|----------------------------------|----------------------------------|
| • Logic for CS | • Probability for CS |
| • Linear Algebra and ODE | • Discrete Mathematics |
| • Fundamentals of Computing | • Data Structures and Algorithms |
| • Computer Organization | • Operating Systems* |
| • Software DevOps | • Introduction to ML* |
| • Multivariate Calculus | • Theory of Computation* |
| • Deep Learning Specialization** | • Programming with Java Script** |
- * - Ongoing ** - Online

EXTRA CURRICULARS

- Volunteer** Dec'22 - Mar'23
Student Placement Office, IIT Kanpur
- Inter-IIT Tech Meet 11.0** Feb'23 - Feb'23
Participated in Inter College Technical Meet in Cyber Security Problem Statement
- Takneek'23** Mar'23 - Mar'23
Mentored a team of 10 students in Inter-Hall Competitions